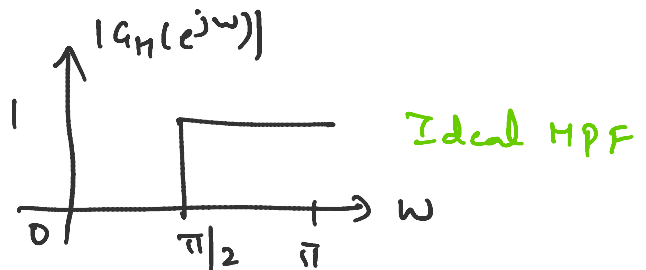
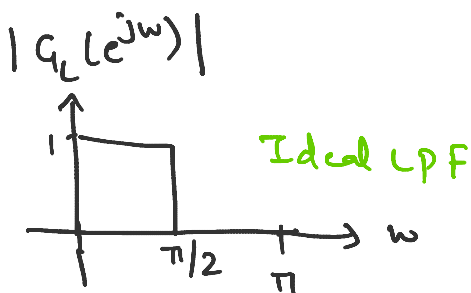
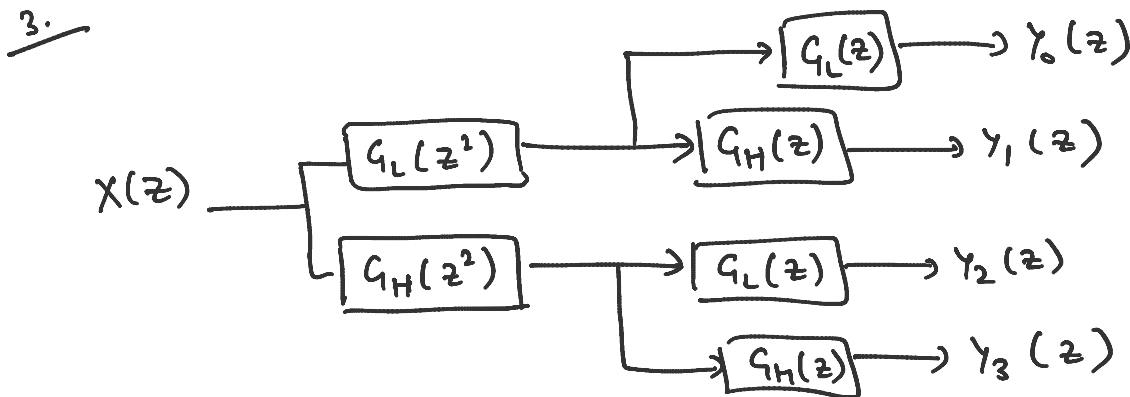
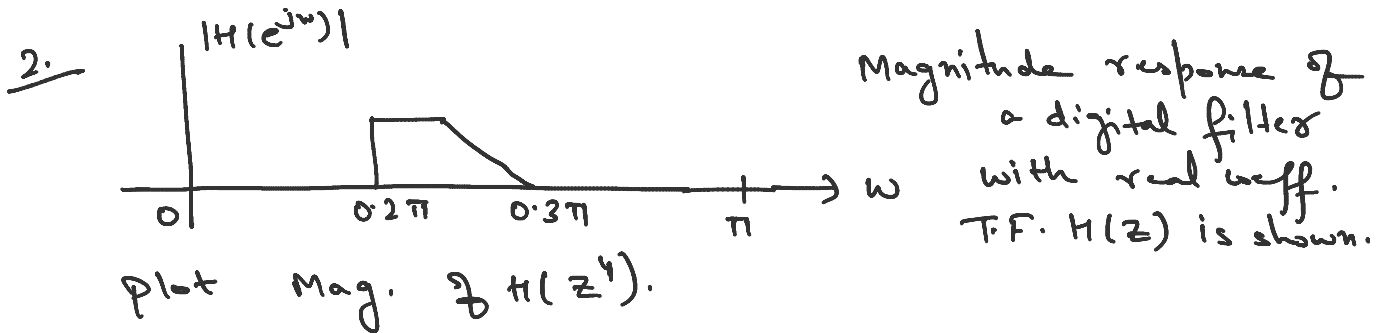


1. A Chebyshev filter is to meet following specifications
 $A_p = 1 \text{ dB}$ for $\Omega = 4 \text{ rad/sec}$
 $A_s \geq 20 \text{ dB}$ for $\Omega \geq 8 \text{ rad/sec}$. Find the T.F. of filter.

Soln: $H(s) = \frac{31.4436}{s^3 + 3.9534 s^2 + 19.814 s + 31.4436}$



Find T.F. of discrete time system given $H_k(z) = \frac{Y_k(z)}{X_k(z)}$,
 $k = 0, 1, 2, 3$, sketch their
 Mag. responses.

4. A third order FIR filter has TF

$$G_1(z) = (6 - z^{-1} - 12z^{-2})(2 + 5z^{-1})$$

- (a) Find all T.F.'s whose mag. responses are identical to $G_1(z)$
 (b) Which one of these filters have min. phase TF
 & which one has max^m phase T.F.

5. Realize IIR TFs in Gray-Markel form, check stability also.

(a) $H(z) = \frac{1 + 2z^{-1} + 3z^{-2}}{1 - z^{-1} + 0.25z^{-2}}$

(b) $H(z) = \frac{1 + 1.6z^{-1} + 0.6z^{-2}}{1 - z^{-1} - 0.25z^{-2} + 0.25z^{-3}}$

(c) $H(z) = \frac{3 + 1.5z^{-1} + z^{-2} + 0.5z^{-3}}{1 - 1.833z^{-1} + 1.5z^{-2} - 0.5833z^{-3} + 0.08334z^{-4}}$

Soln

(a) $k_1 = 0.25, k_2 = -0.8; \alpha_1 = 3, \alpha_2 = 5, \alpha_3 = 4.25$

(c) $\alpha_1 = 0, \alpha_2 = 0.5, \alpha_3 = 1.8986, \alpha_4 = 3.6060, \alpha_5 = 7.8460$
 $k_1 = -0.8444, k_2 = 0.7456, k_3 = 0.4336, k_4 = 0.0833$

6. Develop a 2-Multiplier Cannonic realization of foll.

(a) $H_1(z) = \frac{(1 - \alpha_1 + \alpha_2)(1 + z^{-1})^2}{1 - \alpha_1 z^{-1} + \alpha_2 z^{-2}}$
 (b) $H_2(z) = \frac{(1 - \alpha_2)(1 - z^{-2})}{1 - \alpha_1 z^{-1} + \alpha_2 z^{-2}}$

} α_1 & $-\alpha_2$ are multiplier coeff.

Hint: form difference eqⁿs & then realize the structures.

Hint: form difference eqⁿs & then realize the structures.

7. Develop a realization of 1st order complex coeff. T.F.

$$H(z) = \frac{A + jB}{1 + (\alpha + j\beta)z^{-1}} ; A, B, \alpha, \beta \text{ real constants.}$$

Hint: In H/W or s/w NO 'j' $\therefore$ separate into real & img. parts & realize separately $\rightarrow$ Requires double hardware.

$$\Rightarrow j = \sqrt{-1} \text{ [complex]}$$

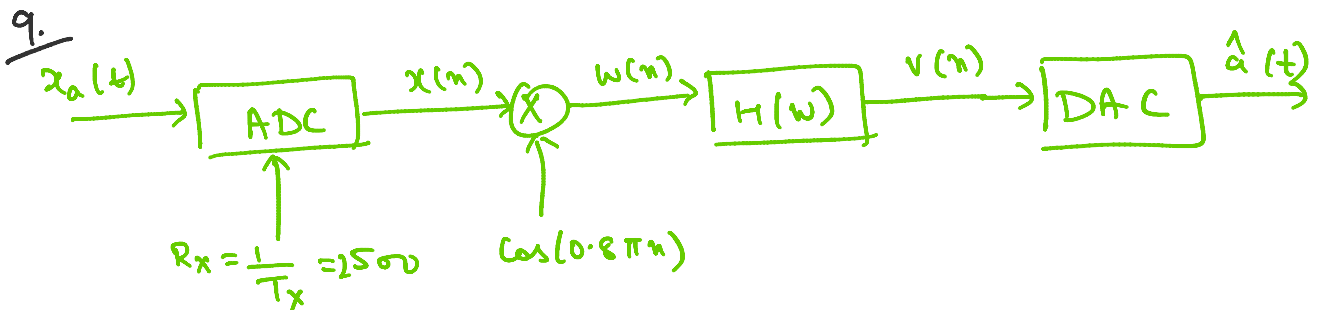
8. Realize as parallel connection of two APFs.

$$(a) H_1(z) = \frac{2 + 2z^{-1}}{3 + z^{-1}}$$

$$(b) H_2(z) = \frac{1 - z^{-1}}{4 + 2z^{-1}}$$

$$(c) H_3(z) = \frac{1 - z^{-2}}{4 + 2z^{-1} + 2z^{-2}}$$

$$(d) H_4(z) = \frac{3 + 9z^{-1} + 9z^{-2} + 3z^{-3}}{12 + 10z^{-1} + 2z^{-2}}$$



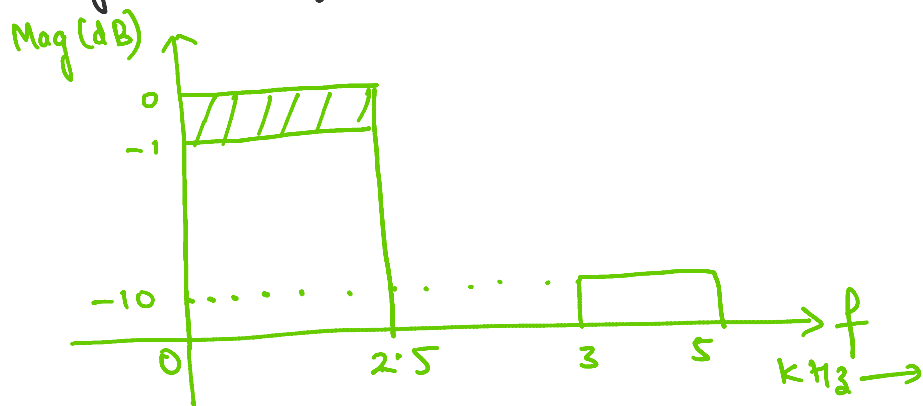
$$x_a(t) = a(t) \cos(2000\pi t), \text{ bandlimited to } 900 \leq F \leq 1100 \text{ MHz.}$$

(a) Determine & sketch spectra for $x(n)$ & $w(n)$

(b) Use Hamming Window of length $M = 31$ to design a LPF that passes $\{a(n)\}$.

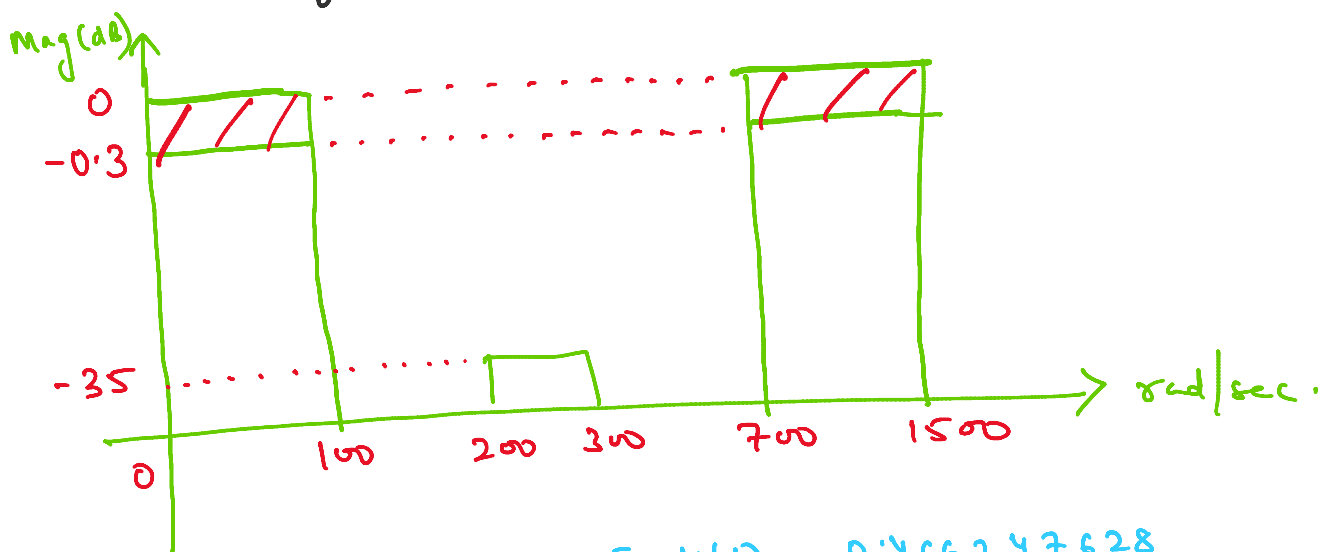
- (b) Use Hamming window of length $M=51$ to design an FIR filter $H(w)$ that passes $\{a(n)\}$.
- (c) Determine sampling rate of ADC that would allow us to eliminate freq. conversion. Soln. 1000

10. Design a Chebyshev LPF using BLT, given foll. specifications



Soln. $H(z) = \frac{0.491(1+z^{-1})^3}{3.7177(1-0.343z^{-1}+0.604z^{-2}-0.204z^{-3})}$

11. Given the filter specifications design a Chebyshev BSF using BLT.



Soln. order = 4, intermediary TF $H_A(s) = \frac{0.466247628}{(s^2 + 0.405s + 1.133)(s^2 + 0.977s + 0.726)}$

————— X —————